KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2015
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

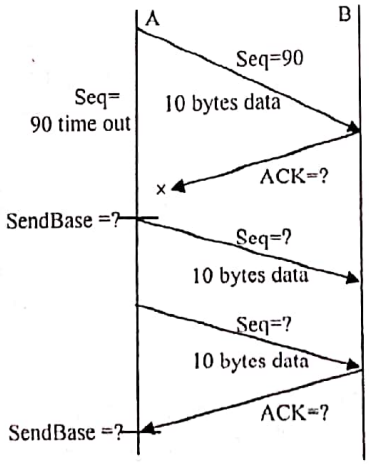
1. a) Define computer network and Internet. Explain Internet on service view points. (09)
   b) Why is it said that packet switching employs statistical multiplexing? Contrast statistical multiplexing with the multiplexing that takes place in TDM and FDM. (09)
   c) "Message segmentation reduces the end to end delay increasing resource usabilities"- Justify the statement. (08)
   d) Explain Internet structure with different tiers. What is the benefit of tierwise ISP connections for internet service? (09)

2. a) Why does TCP socket identification need 4-tuple instead of 2-tuple? What are those tuples? (08)
   b) Why UDP is called a bare-bone protocol in compare of TCP? (06)
   c) In respect of Rdt 2.0, briefly state the objective of the following used techniques: (i) Checksum (ii) ACK packet (iii) NAK packet. (09)
   d) Consider sending a file of 10KB packet with 1 Gbps bandwidth link and 15 msec end-to-end propagation delay. Answer the following questions: (12)
      (i) What is the required transmission time?
      (ii) What is the channel utilization (U)?

Note that $U = \dfrac{\frac{L}{R}}{RTT + \frac{L}{R}}$

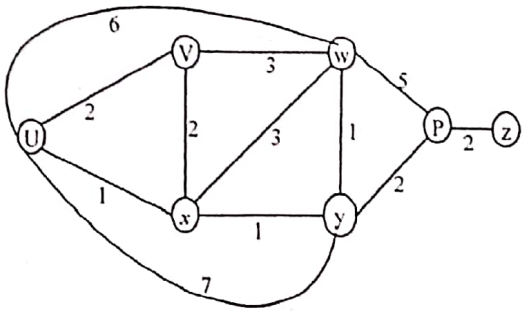
(iii) For getting the channel utilization 0.5 what will be the transmission time?

3. a) What is the performance problem of Rdt 3.0 (stop-and-wait) protocol? How this problem can be resolved? (07)
   b) Sketch a diagram where slective repeat protocol experiences a dilemma. Give your opinion how this dilemma can be overcome. (10)
   c) Consider TCP's communication scenario between A and B. A and B have initial sequence number 90 and 180 respectively. A is sending a file to B with a packet size of 10 bytes in the following figure. Determine the values of seq#, ACK# and sendBase#. (08)

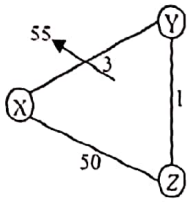


   d) What do you mean by NAT? Explain how NAT operation is performed. (10)

4. a) What is the HOL blocking in the switch fabric of a router? (05)
   b) Consider the following graph. Using Link-state algorithm find the shortest-path tree for router U. (12)



   c) Consider the following graph. Assume that link (x,y) cost has been changed from 3 to 55. Show that the poisoned reverse technique can resolve the count to infinity problem in this case. (11)



   d) Is RIP Intra-AS or Inter-AS routing protocol? Show a simple example of RIP routing table entries. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Briefly explain Nonpersistent HTTP and Persistent HTTP with/without pipeline. Compare TCP connections among the three HTTP methods for a HTML file containing 10 other objects' link. (10)
   b) Differentiate GET and conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cache mechanism? (08)
   c) Does web browsing require UDP service along with TCP? Explain briefly. (05)
   d) "Email demands both push and pull operations"-Justify the statement. (06)
   e) Compare SMTP, HTTP and DNS on the basis of direct transfer and transfer through intermediate hosts. (06)

6. a) Explain common FTP commands and replies. (07)
   b) Explain importance of DNS in service view points. Also explain iterative and recursive DNS queries with appropriate example. (10)
   c) How DNS and CDN together and alone provide better web browsing facilities? (08)
   d) Explain importance of Network Operations Communications Center (NOCC) in content Distribution Network, (05)
   e) "In computer networking switch is transparent to hosts but router is not"-is it true? Justify your answer. (05)

7. a) Why a network interface card is a marked as semi autonomous device? (05)
   b) Explain ideal Multiple Access (MAC) protocol scenario. Compare three different MAC protocols. (09)
   c) Distinguish slotted ALOHA and pure ALOHA protocols. (06)
   d) Write filtering/forwarding algorithm of a switch. List the actions with corresponding condition(s) that a switch may perform when it receives a frame. (09)
   e) Compare routing switches and hubs on operational viewpoints. (06)

8. a) Why does a computer need an MAC address, inspite of having an IP address? (05)
   b) "MAC address of a host never goes outside the network"-is it true? Justify your answer. (05)
   c) Briefly explain various desirable properties of secure communication. (06)
   d) What are shortcomings of symmetric key cryptography? How are those solved in public key cryptography? (07)
   e) What do you mean by digital signature? How does RSA provides digital signature mechanism? (12)